Suraj Gavhale, November 9, 2026.
Virtual Institute of Astroparticle Physics, Paris 75018, France

AI AND THE WORK OF UNDERSTANDING

Physics is not in crisis because its tools are becoming more powerful. AI is expanding what physicists and mathematicians can attempt, that is, searching larger spaces, testing more possibilities and reducing the distance between an idea and a result. The opportunity is to give fundamental questions more of our attention. I am fortunate to be close to many people developing these beautiful systems. Their work is changing the practical conditions of research, and I have considerable admiration for it.

Start with the machinery. In modern autoregressive language models, text and code become tokens represented by numerical vectors. Transformer layers use attention to weight information from the preceding context and pretraining adjusts the model's parameters through gradient based optimization to improve next token prediction. **Ashish Vaswani, Noam Shazeer and their colleagues** [1] gave the foundational Transformer architecture. This training can support sophisticated behaviour, but **predicting a continuation doesn't certify its truth** [2]. Further, **DeepMind's AlphaProof** [3] combines learned proof search with reinforcement learning and Lean, which checks formal proofs. **AlphaEvolve** has language models propose programs, evaluates them and uses an evolutionary process to develop promising candidates [4]. These are mechanisms for producing and testing ideas and their success depends on the problem, the evaluator and the resources available. **Geordie Williamson, Marc Lackenby and András Juhász collaborated with a DeepMind team including Alex Davies, Demis Hassabis and Pushmeet Kohli** [5] to use learned patterns and attribution techniques to guide conjectures and proofs in representation theory and knot theory. AI contributed to mathematical discovery itself.

In physics, **Silviu-Marian Udrescu and Max Tegmark's** [6] AI Feynman used neural network fits, symmetries and separability to simplify the search for equations, recovering known laws from benchmark data. At LIGO, **Jonas Buchli, Brendan Tracey, Rana Adhikari and colleagues** [7] trained reinforcement learning controllers in simulation and tested them on the instrument. They reduced noise from a particular mirror control loop by more than thirtyfold in the *10–30* Hz band. That is a concrete improvement to the means of observation. Like how astronomers used telescopes, scientists can use AI (**Abraham (Avi) Loeb** [8] motivated me to this analogy). An instrument *extends our reach without defining our entire profession.* AI goes further than a telescope in proposing and searching for solutions, but its scientific value still depends on what those solutions establish. This is why physicists need not become merely prompt engineers. Prompt engineering is a useful specialism. Choosing a representation, identifying a hidden assumption, designing a discriminating experiment and explaining a result remain substantive scientific work. A proof checker establishes a formal claim relative to its assumptions. We must still establish that the formal claim captures the intended question and in physics, we must also confront the model with nature. **Terence (Terry) Tao's** [9] essay makes a crucial point. Mathematical progress includes understanding, explanation, theory building and the transmission of knowledge, alongside solving problems. **Jeremy Avigad** [10] similarly argues for pursuing harder questions and broader conceptual ambitions. Their arguments remain valuable even as AI becomes more capable.

Yes, AI takes over parts of the work, including intellectually demanding parts. *That is why it is useful.* Researchers can redirect effort towards fundamental structures, sharper questions and ideas previously too costly to explore. Physicists and mathematicians remain responsible for the substance of their disciplines. There is plenty of serious work ahead. Let us use the tools well, and get on with it.

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